Argo CD GitOps Delivery

Data flow diagram sheets | Revision 1.0 | Sheet 1 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
Boundary labels identify crossed trust boundaries defined in the companion package.

External entity | Process | Data store | Dashed frame = flow grouping

DF1 Boundary TB1

EE1 Administrator workstation → SSH via Internet Gateway; administrator key authentication → P1 SSH bastion

DF2 Boundary TB2,TB3

P1 SSH bastion → Tunneled HTTPS for Argo CLI UI and kubectl configuration → P2 Argo CD API Server; P9 Ops Kubernetes API

DF3 Boundary TB2,TB4

EE2 Application team developer → Authorized developer Git pushes through approved private access; HTTP → P8 Gitea server

DF4 Boundary none

P8 Gitea server → Repository objects permissions and accounts; local persistence → DS3 Gitea repositories and account data

DF5 Boundary TB3,TB4

P3 Repository Server → Git fetch public anonymously or app2 authenticated; HTTP → P8 Gitea server

DF6 Boundary none

P2 Argo CD API Server; P4 Application Controller → Manifest generation or repository requests and results; TLS gRPC → P3 Repository Server

DF7 Boundary none

P2,P3,P4 | See component inventory → Rendered manifest cache reads and writes; Redis TCP without TLS or auth → DS1 Redis rendered manifest cache

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.

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Data flow diagram sheets | Revision 1.0 | Sheet 2 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
Boundary labels identify crossed trust boundaries defined in the companion package.

External entity | Process | Data store | Dashed frame = flow grouping

DF8 Boundary TB3

P2,P4,P5,P6 | See component inventory

Resource watches configuration reads and permitted writes; authenticated HTTPS

P9 Ops Kubernetes API

DF9 Boundary none

P9 Ops Kubernetes API

Persist API objects and retrieve records; logical control plane storage

DS2 Ops Kubernetes persisted Secrets ConfigMaps and custom resources

DF10 Boundary TB2,TB5,TB8

P4 Application Controller

Read live state and apply manifests; authenticated HTTPS cluster wide

EE3 Tenant 1 Kubernetes API app1 app2 cluster wide credential

DF11 Boundary TB2,TB6,TB9

P4 Application Controller

Read live state and apply app3 manifests; authenticated HTTPS namespace scoped

EE4 Tenant 2 Kubernetes API app3 namespace credential

DF12 Boundary TB2,TB7,TB10

P4 Application Controller

Read live state and apply app4 manifests; authenticated HTTPS namespace scoped

EE5 Tenant 3 Kubernetes API app4 namespace credential

DF13 Boundary TB2,TB11

P6 Notifications Controller

Application status notification using service credentials; SMTP with TLS

EE6 Notification email service

DF14 Boundary TB2,TB5,TB6,TB7

P1 SSH bastion

Administrator kubectl over HTTPS through bastion access

EE3,EE4,EE5 | See component inventory

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.

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Data flow diagram sheets | Revision 1.0 | Sheet 3 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
Boundary labels identify crossed trust boundaries defined in the companion package.

External entity Process Data store Dashed frame = flow grouping

DF15 Boundary none

P2,P3,P4,P5,P6,P8,P9 | See component inventory

Application repository and Kubernetes events; local log writes

DS4 Local Argo Gitea and Kubernetes logs

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.